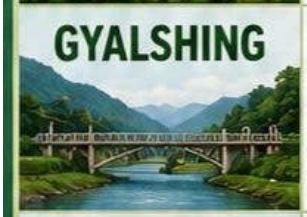
E-WASTE TO WEALTH

RECOVER • REUSE • RECYCLE • REGENERATE

SIKKIM

District-wise E-Waste, Recoverable Metals & Recommended Extraction Methods in Sikkim

DISTRICT	MAIN E-WASTE SOURCES	RECOVERABLE METALS	RECOMMENDED EXTRACTION METHODS
GANGTOK 	IT offices, households, telecom, educational institutes 	Cu, Au, Ag, Pd, Sn, Pb, Ni, Co, Li, Ta, Nb, Ga, In, Nd, Dy, Pr	Mechanical separation; Hydrometallurgy; Electrorefining; Hydrogen Decrepitation; Molten Salt Electrolysis
NAMCHI 	Households, commercial electronics, batteries 	Cu, Al, Fe, Au, Ag, Li, Co, Ni, Sn	Mechanical separation; Hydrometallurgy; Electrowinning
MANGAN 	Consumer electronics, telecom equipment 	Cu, Al, Fe, Au, Ag, Pd, Ta, Nb	Mechanical separation; Hydrometallurgy; Molten Salt Electrolysis
GYALSHING 	Household appliances, LEDs, computers 	Cu, Al, Fe, Au, Ag, Ga, In, Sn	Mechanical separation; Hydrometallurgy; Hydrothermal Processing
PAKYONG 	Airport electronics, communication systems, computers 	Cu, Au, Ag, Pd, Li, Co, Ni, Ta, Nb	Hydrometallurgy; Electrorefining; Molten Salt Electrolysis
SORENG 	Household appliances, mobile phones 	Cu, Al, Fe, Au, Ag, Li, Co	Mechanical separation; Hydrometallurgy
ALL DISTRICTS (where HDDs, EV motors, speakers are collected)	Hard disk drives, speakers, electric motors 	Nd, Dy, Pr (Rare Earth Elements)	Hydrogen Decrepitation → Acid Leaching → Solvent Extraction

RECOVER TODAY FOR A SUSTAINABLE TOMORROW

- Conserves Natural Resources
- Reduces Pollution & Carbon Footprint
- Creates Green Jobs & Local Economy
- Promotes Circular Economy
- Builds a Clean & Green Sikkim

